

Increased Rates of Hip and Spine Fractures in Astronauts are Associated with Longer Spaceflight Durations. Jean Sibonga, Millennia Young, Amy J. Kreykes, Elisabeth Spector and Greg Yardley.

Given the accelerated declines in bone mineral density reported in US astronauts serving on the International Space Station, there is a possibility that a prolonged exposure to spaceflight could put astronauts at increased fracture risk during subsequent planetary explorations or after return to Earth. To date, there have been no fractures in astronauts during spaceflight. Recently-reported data from a flight study suggested that intake of the bisphosphonate alendronate combined with resistive exercise (on the Advanced Resistive Exercise Device – ARED) preserved more astronauts at their preflight skeletal status after a ~6-mos spaceflight than with ARED exercise alone. However, there is low acceptance of alendronate, or the recommended zoledronic acid, to suppress in-flight bone loss largely due to concern for unintended medication side effects and a perceived absence of astronaut fractures that can be directly attributable to spaceflight exposure. Hence, a recent survey of all astronaut fractures was analyzed to assess whether the rate of postflight fractures could be associated with extended spaceflight exposures. The fracture survey was of 262 astronauts across 8433.6 person-years (PY) and grouped according to previous exposure to a “Long Duration (LD)” spaceflight (3 mos or greater, but typically 6-mos). The number of fractures was modeled using count regression adjusting for i) the repeated measures within individuals across the groups, ii) the differing length of surveillance for each individual and iii) age at start of surveillance for each of the groups.

Astronaut Group by Spaceflight Exposure	#Individuals	PY Covered
No Exposure to Spaceflight	262	1882.7
No Exposure to LD spaceflight “Non-LD”	232	6121.3
Exposure to LD spaceflight	42	429.6

Analysis revealed an increased rate of hip and spine fractures for surveillance periods compared between LD Exposure and Non-LD Exposure: Hip – Relative Rate: 3.4 (95%CI: 1.0, 12.0); Spine– Relative Rate: 3.4 (95%CI: 1.0, 11.6). For all other skeletal sites, rates of fracture were similar for LD and Non-LD exposures. Data comparisons were conducted between groups of astronauts to preclude the interpretation of greater physical activity in astronauts as a contributing factor to fracture rates. This analysis of fracture data is the first report associating postflight hip and spine fractures in astronauts with exposure to extended duration spaceflights. Further analyses integrating co-morbidities for bone loss are planned.